

LoFi Sketch: A Large Scale Dataset of Smartphone Low Fidelity Sketches

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ABSTRACT

Recent research on automating the transformation of low fidelity (LoFi) sketches to code using Deep Neural Networks require a large-scale dataset for generalizable results. This paper introduces the LoFi sketch dataset, an open-access dataset of hand-drawn smartphone LoFi sketches, to facilitate this research domain. It contains 4,527 LoFi sketches annotated with 21 categories of UI elements. Through annotations, it provides the category, position, and location of 41,560 constituent UI elements. We collected these sketches from 361 participants from 76 countries through questionnaires and by acquiring premade LoFi sketches to ensure the dataset's diversity and ecological validity. These sketches were drawn using pen, pencil, mouse, touch, and stylus as the input medium and available in raster and vector format. This dataset enables further research on several avenues of AI assistance in UI design, such as UI element detection, sketch completion, UI layout refinement, and Sketch-based image retrieval.

CCS CONCEPTS

• **Human-centered computing** → **Interface design prototyping**; *User interface design*; Wireframes; • **Computing methodologies** → **Image representations**.

KEYWORDS

Dataset, UI sketches, Sketch dataset, Sketch recognition, Low-fidelity prototypes, Wireframes, Sketching interfaces

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1 INTRODUCTION

During the Low fidelity (LoFi) prototyping phase, designers reiterate and rework multiple LoFi prototypes to finalize a User Interface (UI) design. Then, they convert this LoFi prototype to higher fidelities for evaluation and handover to development [3]. This traditional prototyping process is tedious and time-consuming; consequently, it limits the number of LoFi prototype ideas that propagate to higher fidelity. As a solution, several recent research projects endeavoured to generate code from LoFi sketches and wireframes using Deep Neural Networks (DNNs).

These data-hungry neural networks require a large and diverse dataset of LoFi sketches to ensure the generalizability of detection. Therefore, we collected the LoFi Sketch dataset, the first open-access and diverse dataset of 4,527 hand-drawn annotated LoFi sketches of smartphone applications. These sketches were collected from 361 participants, including UI/UX designers, developers, HCI, and CS grad students, from 76 countries. This dataset contains LoFi sketches in both raster and vector formats collected using different input mediums such as pen, pencil, mouse, finger, and stylus. We collected the LoFi sketches in two different ways: collecting sketches through questionnaires and acquiring premade LoFi sketches from participants. This process ensures that the dataset is diverse, representing design-inspiration based sketches and real-life LoFi sketches.

This dataset enables further research on supporting UI designers during prototyping on creating and training DNN models for converting LoFi sketches to code and auto-completing LoFi sketches.

2 RELATED WORK

All DNN based Artificial Intelligence (AI) support for UI design research utilizes either custom datasets or open-source UI sketch datasets. However, most of these research projects do not publish the dataset used for training as open-access. This section looks at the widely used open-source datasets in the current literature on adding AI support in UI design. We list the datasets discussed in this section and their attributes in table 1.

The most widely and largest repository of UI datasets is the RICO dataset [2]. It contains more than 72,000 Android smartphone UI screenshots and its corresponding Android View hierarchy collected using a crowd-sourcing methodology. RICO was further extended by Liu *et al.* by adding semantic annotations for the UI screenshots [7]. The authors categorized the UI elements present in

Table 1: List of UI datasets and their attributes

Dataset	Type	Count	Annotations
RICO [2]	LoFi UI wireframes and UI screenshots	72,000	25 categories of UI elements
Microsoft AI lab [8]	LoFi sketch of Web UI	149	8 categories of UI elements
SWIRE [4]	LoFi sketch of Smartphone UI	3,802	None
Sketch-code [1]	Synthetic Web UI generated using CSS	1,750	from HTML Code
Syn [9]	Synthetic LoFi sketch generated by random allocation	125,000	21 categories of UI elements
SynZ [10]	Synthetic LoFi sketch generated from RICO UI layouts	175,377	21 categories of UI elements
UISketch [11]	UI element sketches	17,979	None

the RICO dataset into 25 categories using manual and automated methodologies in their research. Further, they classified these UI element categories into 197 text button concepts and 135 icon classes. One drawback of this dataset is that as the UI elements were categorized and annotated using automated means, the annotations are unreliable. Therefore, using this dataset for training DNNs would cascade the inaccuracies to the trained DNN model. Despite this limitation, this dataset is used in many existing works of literature, typically after a few refinements.

Recently, Leiva et al. [5] revised a subset of the RICO dataset to overcome its shortcomings and provided a refined dataset, Enrico (shorthand of Enhanced Rico). This dataset is annotated with human supervision and contains 1,460 UIs classified into 20 design topics. Enrico is annotated into 25 categories of UI elements similar to RICO.

Microsoft AI labs created a small LoFi sketch dataset to train their proof-of-concept project, Sketch2Code [8]. This dataset contains 149 sketches of web UIs annotated with ten categories of UI elements. Although this dataset is used in a few other research projects, it is insufficient to train a generalizable DNN model due to its small size. Also, the collection method of this dataset is not specified to understand its characteristics.

Another UI dataset of LoFi sketches is the SWIRE dataset [4]. This dataset contains 3,802 LoFi sketches based on 2,201 UI screenshots from the RICO dataset to train their GUI similarity search DNN. SWIRE is an unlabelled, unannotated dataset of smartphone LoFi sketches. These sketches were collected from five UI designers. Although this is the most extensive collection of the LoFi sketch dataset in the existing literature, this is not yet used in any other research other than GUI similarity search.

A few projects in this domain proposed synthetic LoFi sketch datasets such as Sketch-Code [1], Syn [9], and SynZ dataset [10]. Synthetic data is an alternative to real data, commonly used to overcome time, funds, or privacy limitations. However, it cannot replace real data and is used typically for training the base DNN model while relying on real data to fine-tune the model further [10].

A recent addition to the UI datasets is the UISketch dataset [11]. This dataset contains 17,979 hand-drawn sketches of 21 UI element categories collected from 967 participants. The sketches in the dataset were collected from 10 different countries using paper and digital questionnaires. However, these sketches represent individual

UI elements, and therefore they do not accurately depict a real-life LoFi sketch.

3 LOFI SKETCH DATASET

To overcome the limitations of the prior research, we propose collecting a large-scale LoFi sketch dataset from diverse participants with various input mediums. This section describes the steps we followed to ensure this and create the LoFi sketch dataset.

3.1 Design decisions

We scoped this dataset for LoFi sketches of smartphone applications. Based on the data collection practices from prior researches on collecting UI sketches [4, 11], we decided to provide a sample UI screenshot to the participants from the RICO dataset and ask them to sketch it as a LoFi prototype. Therefore, we collected and annotated 100 UI screenshots from the RICO dataset. Additionally, we aimed to collect a diverse dataset representing scenarios where LoFi sketches were made from UI design inspirations and real-life scenarios where LoFi sketches were made. Therefore, to obtain ecologically correct samples, we also collected LoFi sketches of real LoFi prototypes that our participants made in addition to the data collection.

Regarding the medium of sketching, we collected the digital LoFi sketches and paper LoFi sketches. In digital sketches, we allowed participants to sketch the prototype using either stylus, touch, or mouse as input, and in paper-based sketches, we allowed participants to use pen and pencil. With these design decision, we started the data collection by creating paper and digital questionnaires.

3.2 Data collection questionnaires

We created the paper and digital questionnaire similar to prior research work by Pandian et al. [11] to collect the LoFi sketch dataset.

We created the paper questionnaire such that each question has an example of a UI screenshot on the left and a rectangular area to sketch the UI element on the right-hand side (Figure 1(a)). These example images were randomly selected from the 100 UI screenshots we picked from the RICO dataset and intended to assist the participants by providing a UI design inspiration.

We created a web application for the LoFi sketch digital questionnaire. This questionnaire allows mouse, touch and stylus input to sketch LoFi prototypes (Figure 1(b)). The web application starts

by requesting participants to provide informed consent and provides an informative video to understand the purpose of this data collection. Once the participant agrees to the consent form, similar to the earlier versions, this web application creates a secure key for authentication and stores it in their browser's internal database. This step restricts the participants from taking part in the study multiple times. It also helps restore the web application from where the participant left off in case of network disconnection.

The digital questionnaire stores the sketch in vector and raster form. Additionally, it stores temporal information of each stroke made by the participant. This temporal information allows us to extract details about the time taken for each stroke and the overall sketching time of the participant in creating a LoFi sketch.

3.3 Data collection procedure

As we aimed to collect a diverse dataset representing design inspiration based sketches and real-life LoFi sketches, we conducted the data collection procedure in two different ways: collecting sketches through questionnaires, acquiring premade LoFi sketches from participants. Table 2 summarizes the count of LoFi sketches collected by various procedures. In this section, we discuss both the procedures successively.

3.3.1 Participants. We obtained the LoFi sketches from 361 participants: 29 UI/UX designers, 49 developers, 251 HCI-grads, and 32 CS-grad students from 76 different countries. The participants had 1-14 years of prototyping experience (Mean=1.5, S.D=1.54).

3.3.2 Questionnaire-based. The nationwide lockdown in 2020 and 2021 due to COVID-19 limited our data collection process; therefore, we formulated our data collection procedure accordingly. As in-person meetings are restricted, we concentrated on collecting data through digital questionnaires.

The initial steps in the data collection procedure for paper and digital questionnaires were similar. We explained to the participants the reason and benefits of the study and asked them to provide us with their demographics (optional) and consent (mandatory) to participate in the study. We then explained the questionnaire and how to use it. We requested the participants not to trace the

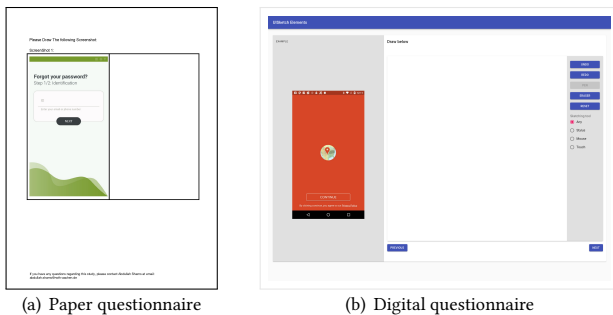


Figure 1: Paper and digital questionnaire used for collecting LoFi sketch dataset. Both questionnaires have an example of a inspiration UI screenshot on the left and a rectangular area to sketch the LoFi on the right-hand side.

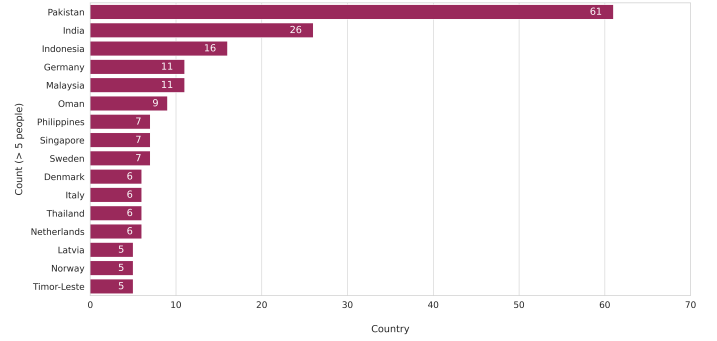


Figure 2: Count of participants (>= 5 people) per country that contributed to LoFi sketch dataset

Table 2: Split of participants and count of collected sketches based on data collection source

	Participants	Count of collected LoFi Sketches
Digital Questionnaire	343	8,860
Paper Questionnaire	18	214
Other sources	-	1,201
Total	361	10,275

UI screenshot example; instead, we recommended using it as an inspiration to model their LoFi sketch based on it. We did not restrict the number of LoFi sketches the user could draw, and then we terminated the study once the participant decided to stop the study. We informed them that they could also resume the study at any later time. We also provided the participants with the flexibility to skip a UI design sample if they were not comfortable sketching it. Although this approach provided us with unequal distribution of LoFi sketches drawn by participants, this allowed the participants from being overwhelmed by the task.

Using this procedure, we collected data from 18 participants (~5%) using paper questionnaires and 343 participants (~95%) using digital questionnaires. In total, we collected 214 paper-based LoFi sketches and 8,860 digital LoFi sketches using the questionnaires.

3.3.3 Other sources. We contacted several universities, research institutes, and companies to acquire LoFi sketches. As LoFi sketching is a conventional process followed before building real-life GUI applications, we requested them to forward LoFi sketches made by UI/UX designers or HCI grad students for their previous projects. However, most companies and research institutes refused to provide their LoFi sketches due to the privacy policy. Few companies and research institutes provided the sketches but without the demographic details. We also acquired a few LoFi sketches students drew for their course capstone projects.

In total, we acquired 695 paper-based LoFi sketches by this process. However, we could not map this information to participants or their demographics due to privacy restrictions. Therefore, this information is not well-represented in the participant information.

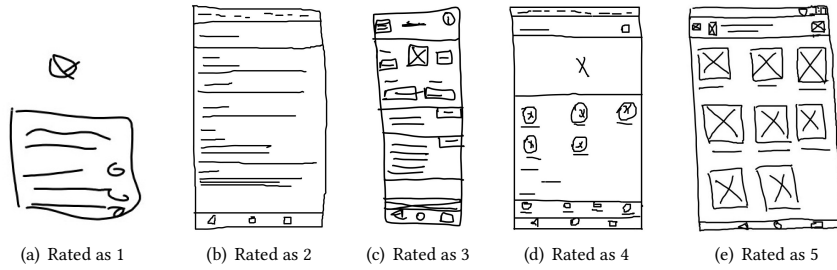


Figure 3: Samples of LoFi sketches rated in a scale of 1 to 5.

Table 3: Overview of sketches removed during LoFi sketch data verification process

Sketches	Count of LoFi sketches
Collected	10,275
With little to no details	3,027
With rating < 5	2,721
LoFi sketch dataset	4,527

3.4 Data verification

In total, we collected 10,275 LoFi sketches from various sources from our data collection process. As a next step, we manually merged all the collected LoFi sketches, reviewed them, and cleaned them.

In digital questionnaires, many participants decided to skip sketching a sample by leaving the sketching area empty or leaving the sketch unfinished. Therefore, we started the data verification process by removing the sketches with little to no detail. In this step, we identified and discarded 3,027 LoFi sketches (29.46%).

Also, as the details present in a LoFi sketch varies due to the input medium (pen, pencil, mouse, touch, and stylus), we added rating to the sketches on a scale of 1 to 5, with one being the worst and five being best. We then discarded all the sketches with a rating of less than 5. Figure 3 shows a sample LoFi sketch with each rating in our scale. By this process, we removed 2,721 LoFi sketches (26.48%). We performed this step to improve the quality of the sketches in the final LoFi sketch dataset.

Through the data verification steps described above, we removed 5,748 LoFi sketches (55.94%) from the 10,275 LoFi sketches collected from various sources and finally accepted 4,527 sketches in the LoFi sketch dataset. Table 3 summarizes the count of LoFi sketches collected and removed by the data verification process.

3.5 Data annotation

Our goal for creating the LoFi sketch dataset is to provide a large-scale annotated dataset for training DNN models. Therefore, after the data verification process, we created annotations for each LoFi sketch in the dataset.

A LoFi sketch comprises various UI elements present in different locations and has different dimensions. Through this dataset, we aimed to capture the information of the constituent UI elements in a LoFi sketch along with their location and dimension in that sketch.

To annotate these sketches, we used the Supervisely¹ platform. We loaded the 4,527 LoFi sketches in Supervisely and annotated them into the 21 UI element categories as classified in the UISketch dataset [11]. The annotation was done with the help of four annotators and reviewed by two UI/UX researchers. Figure 4 shows a sample LoFi sketch with its annotations.

3.6 Collected data

With the annotation files, we created the LoFi sketch dataset. This dataset contains 4,527 LoFi sketches annotated with 41,560 constituent UI elements classified into 21 categories. Most LoFi sketches were collected from paper and digital questionnaires using pen, pencil, mouse, touch, and stylus as the input medium. On average, each LoFi sketch contains 9.18 UI elements (SD=5.72, Median=8). For the LoFi sketches collected using a digital questionnaire, we measured the time taken by the participant to sketch a LoFi. On average, a participant took 52.93 seconds to sketch an interface. We have included the sketch type, time taken in seconds, and input method information for each LoFi sketch, along with the annotations.

We provide this dataset with the LoFi sketches and their respective annotation files as open-access for further research in this domain². Figure 5 visualizes the distribution of LoFi sketches collected using the questionnaires based on the medium.

¹<https://supervise.ly/>

²<https://www.kaggle.com/vinothpandian/lofisketch>

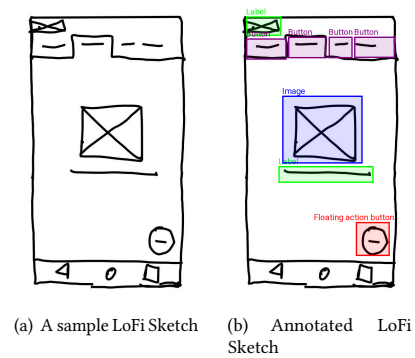


Figure 4: Sample LoFi Sketch with its annotations

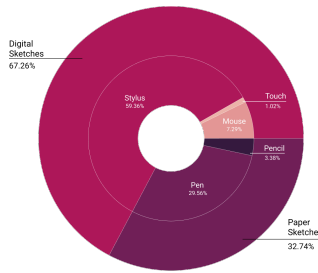


Figure 5: Distribution of digitally drawn LoFi sketches using stylus, mouse, and touch vs. paper sketches drawn using pen and pencil in LoFi sketch dataset

4 BENEFITS & APPLICATIONS

LoFi sketch is the first large-scale annotated LoFi sketches dataset. This dataset opens up several possible research areas.

UI element detection. Many research projects have attempted to convert LoFi sketches to code using DNN models. Although this dataset is not enough as stand-alone to train such DNN models, it can be combined with synthetic datasets such as Syn and SynZ datasets to create generalizable UI element detection networks.

Sketch Completion. Li et al. [6] explored a novel research area of auto-completing UI wireframes. This dataset contains rich information about UI element categories and their layout in a LoFi sketch; therefore, it can be utilized for training auto-completion DNNs which supports designers by autocompleting their strokes.

UI layout refinement. The 4,527 annotations in the LoFi sketch dataset contain information about the position and dimension of 21 categories of UI elements. This position and dimension data can be utilized for training UI layout refinement DNNs. UI layout refinement reduces the designers' manual alignment process and enhances UI layout aesthetic using AI.

Sketch-based Image Retrieval. As the LoFi sketches in this dataset were collected by diverse participants using various input mediums, this dataset can be utilized to fine-tune sketch-based image retrieval (SBIR) proposed by Huang et al. [4]. This further fine-tuning will improve the generalizability of the SBIR model.

5 SUMMARY & FUTURE WORK

This paper introduced the LoFi sketch dataset, the first open-access dataset of annotated LoFi sketches containing 4,527 LoFi sketches annotated with 41,560 constituent UI elements. We collected these sketches from 361 participants from 76 countries through questionnaires and by acquiring premade LoFi sketches from participants. It contains sketches in both raster and vector format sketched by participants using pen, pencil, mouse, touch, and stylus as the input medium. This dataset enables further research on supporting UI designers during the prototyping process by creating and training DNN models for converting LoFi sketches to code and auto-completing LoFi sketches. As future work, we aim to collect sketches for other platforms such as web and desktop applications. We also aim to improve the LoFi sketch annotations by adding contextual information such as the handwritten text and the stroke's

colour. Further, using the LoFi sketches from this dataset, we aim to analyze how designers understand and communicate UI design and compare their UI element detection accuracy with DNN based object detection models.

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